Philipp Nazari

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Education

PhD **Max Planck ETH Center for Learning and Systems**, Computer Science — July 2025 – July 2028
- CLS PhD Fellow, advised by Konstantin Rusch and Prof. Fanny Yang
- Research Areas: Efficient ML across the full stack — architecture design, post-training compression, adaptive inference

MsC **ETH Zurich**, Mathematics — Sept 2023 – Feb 2025
- GPA: 5.78 (Swiss grading system)
- Focus Areas: Differential Geometry, Machine Learning

BsC **Ruprecht-Karls-University Heidelberg**, Mathematics — Oct 2019 – July 2023
- GPA: 1.3 (German grading system)
- Focus areas: Differential Geometry, Analysis, Machine Learning

University of Bergen, Mathematics — Aug 2021 – Jan 2022
- Exchange Semester at the University of Bergen, Norway
- Focus Areas: Algebraic Topology, Differential Geometry

BsC **Ruprecht-Karls-University Heidelberg**, Physics — Oct 2019 – Aug 2022
- GPA: 1.3 (German grading system)
- Focus areas: Theoretical Physics, Machine Learning

Experience

Heidelberg Collaboratory for Image Processing (HCI), Research Assistant — Heidelberg, Germany
Aug 2022 – Mar 2023

Publications

On State Reduction in Linear Attention — June 2026
Philipp Nazari, T. Konstantin Rusch
openreview.net/forum?id=KkN6enKAlU (***Oral*** AdaptFM workshop at ICML 2026)

The Key to State Reduction in Linear Attention: A Rank-based Perspective — February 2026
Philipp Nazari, T. Konstantin Rusch
arxiv.org/abs/2602.04852 (Preprint)

The Curious Case of In-Training Compression of State Space Models — October 2025
Makram Chahine, ***Philipp Nazari***, Daniela Rus, T. Konstantin Rusch
openreview.net/forum?id=LtzmeSMBTW (ICLR 2026)

Geometric Autoencoders – What You See is What You Decode — July 2023
Philipp Nazari, Sebastian Damrich, Fred Hamprecht
proceedings.mlr.press/v202/nazari23a.html (ICML 2023)

Software

Discretax: State Space Models in JAX — October 2025
Philipp Nazari, Francesco Maria Ruscio, Benedict Armstrong, T. Konstantin Rusch
github.com/camail-official/discretax

Research Projects

Geometric Encoder Regularization in Autoencoders Juli 2024

- Semester Project at ETH with Prof. Thomas Hofmann. Paper and code

Entropy Aware Message Passing in Graph Neural Networks Mai 2024

- Project spun out of the Deep Learning course at ETH by Prof. Thomas Hofmann. Paper and code

Talks

Theoretical Perspectives on Efficient Architectures June 2026

- Modern Numerical Methods for Theoretical Physics seminar, Heidelberg University

On State Reduction of State Space Models May 2026

- Tübingen AI Center. Recording

An Introduction to Autoencoders and (Geometric) Regularization Techniques November 2024

- Machine Learning and Physics course, Heidelberg University. Recording: part 1, part 2

Press

New technique makes AI models leaner, faster while still learning April 2026

- Coverage of our work on in-training compression of state space models in MIT News